The Association Between Moderate Physical Activity and Self-Reported Depression Outcomes

Evidence from the National Health Interview Survey, 2010–2023

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Paper in press –
University of Michigan Undergraduate Journal of Public Health

BACKGROUND

21M
U.S. adults had a major depressive episode in 2021 (8.3%; NIMH)

280M
people live with depression worldwide (WHO)

17.2%
of U.S. adults aged 18–25 had depression in 2020, up from 10.3% in 2015 (Goodwin et al., 2022)

Treatment barriers
- High costs
- Long waitlists
- Provider shortages

→

Physical activity
- Low-cost
- No prescription
- Flexible, anywhere

Prior evidence: activity is linked to lower depression risk

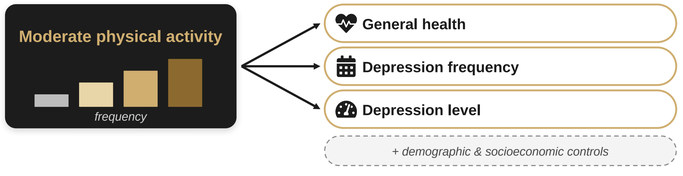
Any activity *Kandola et al., 2019*	**17% lower odds**
Half the recommended dose *Pearce et al., 2022*	**18% lower risk**
Full recommended dose *Pearce et al., 2022*	**25% lower risk**

The gap: few nationally representative studies compare how often people exercise across several dimensions of depression.

AIMS

Q **Question:** Is how often U.S. adults do moderate physical activity linked to their self-reported depression and general health?

H **Hypothesis:** Since activity can lower stress and aid sleep, more frequent activity is linked to better depression outcomes.



METHODS

Data: IPUMS NHIS, 2010–2023

N = 332,521 U.S. adults · self-reported · nationally representative

Exposure: moderate activity, times/week

None 0 | **Low** 1–3 | **Medium** 4–6 | **High** 7+

Outcomes: 3 self-reported measures

General health | Dep. frequency | Dep. level

Model: OLS regression vs. no activity

(3) + Socioeconomic: marital status, income, poverty ratio, meds ★ *preferred*

(2) + Demographic: age, sex, race

(1) Activity only

How to read: each estimate is the percentage-point difference vs. inactive adults.

RESULTS

−2.92
percentage points

KEY FINDING

Adults active 4–6 times a week were 2.92 percentage points less likely to report frequent depressive symptoms than inactive adults.

95% CI: −4.4 to −1.4 · Specification 3

Figure 1. Change in each outcome vs. no activity, by activity level (Spec 3, 95% CI)

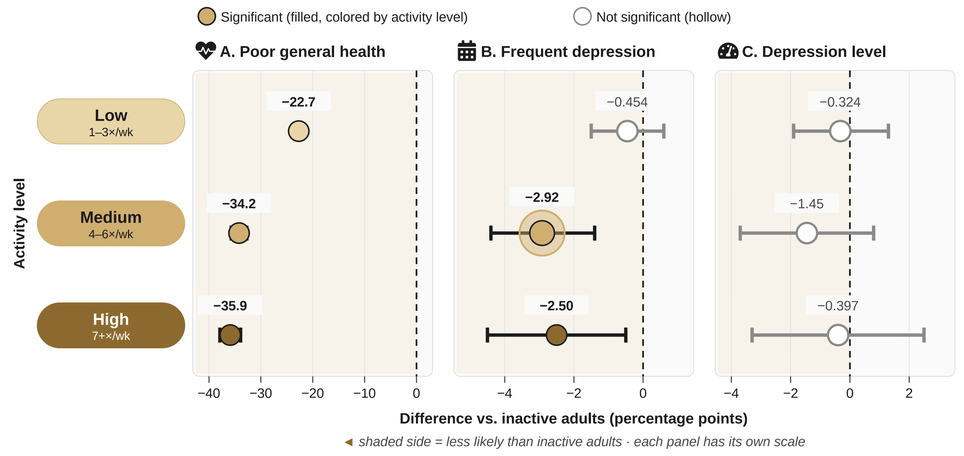


Figure 2. Who was studied: sample by activity level (N = 332,521)

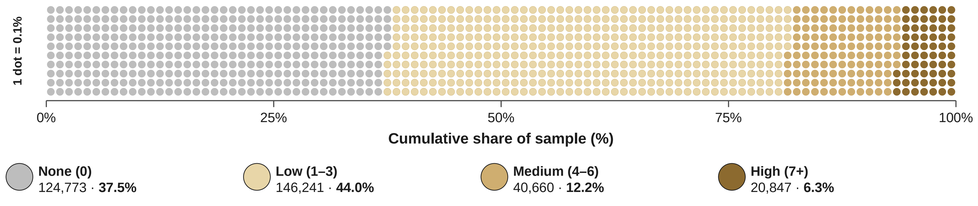


Table 1. Specification 3 estimates, percentage points [95% CI]

Activity level	General health (poor)	Depression frequency	Depression level
Low (1–3)	**−22.7*** [−23.8, −21.6]	−0.454 [−1.5, 0.6]	−0.324 [−1.9, 1.3]
Medium (4–6)	**−34.2*** [−35.7, −32.7]	**−2.92*** [−4.4, −1.4]	−1.45 [−3.7, 0.8]
High (7+)	**−35.9*** [−37.9, −33.9]	**−2.50*** [−4.5, −0.5]	−0.397 [−3.3, 2.5]
Observations	161,344	160,974	66,089
R^2	0.204	0.252	0.038

** Significant (95% CI excludes 0); highlighted. Reference group: no moderate activity.*

DISCUSSION

- **4–6 sessions a week** had the largest link with fewer frequent depressive symptoms
- **Better self-rated health** at every activity level (−22.7 to −35.9 pp)
- **Fits prior work** (Kandola et al., 2019; Pearce et al., 2022)
- **Socioeconomic status matters:** welfare receipt consistently predicted worse outcomes
- **No link with depression level:** all CIs include 0 (R^2 = 0.038)

Limitations

- **Self-report bias**
- **Unmeasured confounders**
- **Association, not cause**
- **Income only in categories**

CONCLUSIONS

1. **Being moderately active 4–6 times a week was linked to a lower chance of feeling depressed often.**
2. Any regular moderate activity – even 1–3 times a week – was linked to better self-rated health.
3. Activity was not linked to how strongly depression is felt, so it is a low-cost complement to care, not a cure.

Big picture: as depression rises and care stays hard to reach, regular moderate activity is a low-cost tool worth promoting.

Future directions

Test activity as a treatment for people clinically diagnosed with depression, tracking real-time, individual-level change.

ACKNOWLEDGEMENTS

I thank Professor Osborne for his helpful feedback on the manuscript, and my mom and dad for their constant support and encouragement.

Funding: This independent study received no external funding.
Data: IPUMS NHIS v8.1 (Blewett et al., 2025).

Author Contribution Statement

NG (sole author) conceived and designed the study, prepared the data, performed all analyses, interpreted the results, wrote the manuscript, and designed this poster.

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